COCONUT RESEARCH INSTITUTE OF SRI LANKA

Advisory Circular No A 10

CULTIVATION OF COVER CROPS IN COCONUT LANDS

One of the most effective methods for conserving soil moisture in coconut lands is to establish a cover crop.

Benefits of cover crops

- Cover crops reduce surface run-off and erosion by breaking the impact of rain drops.
- It provides a large quantity of organic matter to the soil.
- It improves the soil structure and infiltration of water by enhancing the conservation of moisture.
- It reduces leaching of nutrients, soil temperature and weed growth
- Leguminous cover crops provide nitrogen to the soil
- Cover crops are essentially surface feeders and generally do not compete for moisture with relatively deep-rooted plantation crops.

Characteristics of a suitable cover crop

- It should grow quickly and cover the soil in a short period of time, thereby controlling weeds
- It should dry up during drought and should not compete for moisture. It regenerate during rains and also produce seeds
- It should tolerate shade and provide a large quantity of mulch

Recommended cover crops

A cover crop should be selected after considering the climate of the area, soil conditions and the shade in the land. In the wet zone areas under heavy rainfall (1,875 to 2,500 mm), cover crops are useful against soil erosion and improving the soil fertility by adding organic matter. In the intermediate zone (1,000-1,875 mm rainfall) and dry zone(less than 1,000 mm rainfall), cover crops are drying up during the dry months, adding large quantities of leaf litter and then regenerate with the onset of rains.

Creeping cover crops

For wet areas: (Wet and wet intermediate zones)

1. *Pueraria phaseoloides* ('Puero')
2. *Calopogonium mucunoides* ('Calopo')
3. *Centrosema pubescens* ('Centro')

For dry areas : (Dry intermediate and dry zone)

1. *Centrosema pubescens* ('Centro')
2. *Macroptilium atropurpureum* ('Macro')
3. *Pueraria phaseoloides* (for dry intermediate zone only)
4. *Calopogonium mucunoides* (for dry intermediate zone only)

Mucuna utils ('Wanduru-me'), which covers the ground rapidly within 3-4 months is a good cover to plant in infertile soils quick rehabilitation of land, particularly before replanting is done.

Bush cover crops

1. *Gliricidia sepium*
2. *Gliricidia maculate* ('Weta-mara')

Gliricidia is suitable for any soil type and climate.

When the growers plan to establish cover crops, they should make the necessary arrangements, particularly ordering seeds, well in advance.

Some notes on the recommended cover crops are given below.

***Purearia phaseoloides* ('puero')**

Earlier called *Pueraria javanica*.

This is a climbing or creeping perennial, spreading by long vigorous runners and rooting at nodes to provide a deep, smothering mat. Leaves have three large round and hairy leaflets; (figure 1 & 2) flowers with colour varying from blue to pale bluish purple; seed pods cylindrical; pods mature

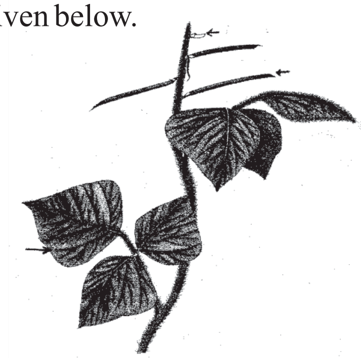

Figure 1: Line diagram of a Pureria phaseoloides cover crop

irregularly; seeds small, oblong, dark brown and susceptible to insect attack and sometimes show poor germination. One kg of seeds contains about 80,000.This cover crop can be grown in a wide range of soils. Although it does not thrive well on sandy soils, it is particularly useful for clay and gravelly soils. It can also be grown in water-logging and moderate shady lands. This species can be established either from cuttings or seeds. It has medium growth rate but within 9 to 12 months it forms a thick ground cover. It dies back during dry periods, but recovers after few showers. It cannot tolerate heavy grazing or cutting.

Figure 2: Pureria phaseoloides cover crop

***Calopogonium mucunodes* ('Colopo')**

This is a climbing or creeping perennial which is in general resembles to puero Leaves have three ovate leaflets but smaller than Puero (figure 3 & 4). Roots at nodes, flowers small and blue color; pods brown and narrow covered with fine brown hairs; seeds flattened, squire in shape and light brown. One kg of seeds contains about 70,000.

This species can be established by seeds or cuttings. It thrives well on a wide range of soils and forms a thick cover in 6 to 8 months. It also tolerates shade to a great extent. It dies back during drought, but regenerates quickly after the rains. Frequent droughts can however kill the cover crop. Cattle are reluctant to graze on this cover crop. This may be cultivated as a mono-crop or Puero and Centro mixed crop.

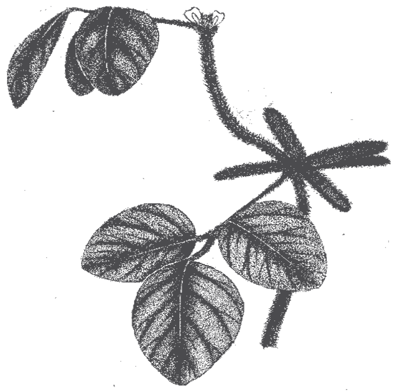

Figure 3: Line diagram of a Calapogonium muconoides cover crop

Figure 4: Calapogonium muconoides cover crop

***Centrosema pubescens* ('Centro')**

This is a climbing perennial with long stems; often rooting at the nodes; leaves small with three oval shaped leaflets with pointed tips (figure 5 & 6). Flowers are in dark purple color; pods are dark brown, long, narrow with a sharp tip. There are about 40,000 seeds per kg.

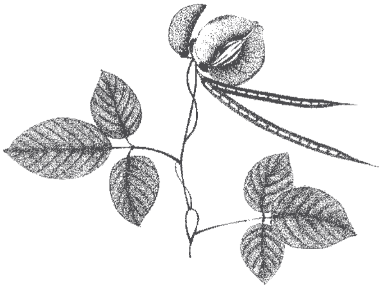

Figure 5: Line diagram of a Centrocema pubesons cover crop

Figure 6: Centrocema pubesons cover crop

It thrives well on a wide range of soils. Growth during early stages is slow and generally slower than Calopo and Puero. It can tolerate poorly drained soil conditions. Seeds are used for the establishment. Although mature Centrosema can tolerate shade, growth rate at early stage is slow. Although it is a very hardy plant which does not completely die during dry season, it adds a large amounts of leaf litter. With the onset of rain it recovers rapid. This species can withstand heavy grazing. Suitable to cultivate in cover crop mixtures with rapidly growing Puero and Centro and also with pasture grasses.

***Macroptilium atropurpureum* (Siratro)**

This species has underground stems, and roots at the nodes (figure 7& 8). Leaves have three broad leaflets, characteristically lobed and green on the upper surface and covered with silvery - gray fine hairs on the lower surface. Flowers deep red; pods narrow, straight and cylindrical, carrying about 12 seeds, shattering at maturity; seeds are dark brown. One kg of seed contains about 80,000. Its natural ability of spreading seeds freely on the ground promotes its spread.

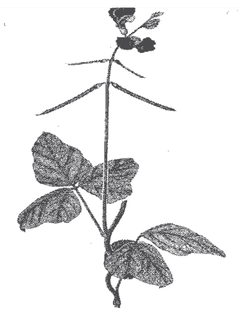

Figure 7: Line diagram of a Macroptilium artoperpurium cover crop

Although it thrives well on a wide range of soils, Siratro prefers light-textured soils. Because of its relatively deep tap root, it can tolerate drought and thrives in dry areas adding considerable amounts of leaf litter. It performs well under moderate shading and also preferred by cattle. It is also suitable to grow with other cover crops and fodder grasses as mixtures.

Gliricidia

Figure 8: Macroptilium artoperpurium cover crop

This is a fast-growing, medium - sized leguminous tree (figure 9). There are two distinct species, namely Gliricidia sepium, which produces pink coloured flowers and large round seeds (about 5,000/kg) and Gliricidia maculata, which produces white-coloured flowers and small round seeds. Of these, *G. sepium* is more common.

It grows rapidly producing a large quantity of foliage which can be used as a green manure. The leaves decompose relatively fast, providing nitrogen, potassium etc. After loping, re-growth of Gliricidia is rapid enabling quick recycling of minerals. This species can be propagated by cuttings and seeds. It grows well in low-fertile soils and tolerates drought. Gliricidia is also a useful source of fuel wood, fence poles and animal feed.

Figure 9: Single row Gliricidia cultivation

Planting and management of creeping covers

Cover crop seeds can be purchased from seed merchants or seeds can be collected from existing fields where cover crops are available.

Seeds

Most cover crops flowers in October/November and mature pods turned into straw-brown colour can be harvested around January/March. When the pods are ready for harvesting, they can be handpicked every other week. A pod contains around 8 - 12 seeds. After harvesting, the pods should be sun-dried for 2-3 days. The pods are then covered with a mat and trampled to obtain the seeds. Good seeds can be separated by winnowing.

Land preparation

The land should be free of weeds to facilitate rapid establishment of the cover crop. The land should first be ploughed and then disc-harrowed across the ploughed furrows to remove weeds. The disc-harrowed surface should be leveled with mammoty forks or hand rakes. When cover crops are planted for the first time and particularly in poor sandy soils or on eroded slopes, it is recommended that a starter application of a small amount of cow dung or a small dose of artificial fertilizer such as a mixture of ½ kg urea, 1-2 kg of saphos phosphate and 1 kg of muriate of potash per coconut square be given.

Planting

It is always advisable to establish cover crops with the south-west monsoon rains (May/June), preferably at the beginning of rains as normally seeds are harvested around March, before sowing, seeds should be dipped in hot water for 3 minutes followed by soaking in cold water for 12 - 24 hours. Seeds so treated must not be allowed to dry before sowing. After sowing, seeds should be covered by a light forking with mammoty forks, hand rakes or by a chain harrow.

1. Germination of seeds in a soil and cow dung mixture placed coconut husk.
2. Germination of seeds in polythine tub filled with a soil and cow dung mixture (figure10)

Seeds can be broadcast in rows 60 cm apart, which would require about 5-6 kg seeds/ha. Seeds could also be broadcast in the entire area (leaving the manure circle) and this would require 8-10 kg seeds/ha. If planted in rows, it is possible to do weeding between rows.

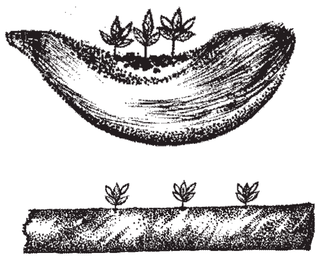

Figure 10: Germination of seeds of cover crops

Cover crops should be conveniently established on husk/coir dust pits or trenches.

If the seeds are expensive or if the supply is limited, they can be germinated in a nursery. A traditional method of raising plants is to germinate the seeds on coconut husk containing a mixture of soil and dung (figure 10. i). Another convenient method is to plant them in long polythene tubes (about 5 cm in diameter) filled with soil/dung mixture. The tubes are laid flat and holes are made at 1' intervals. Seeds are then dibbled in to the soil (figure 10. ii). The seedlings thus raised can be planted in the field at regular intervals (say 1 meter apart or in 3-4 patches/coconut square).

Some cover crops could also be propagated by cutting with adventitious roots. Cover crop species may be planted alone or mixed depending on the soil type and climate zone. It is always advisable to have a mixture or cover crop species to minimize the risk of certain covers being completely wiped out during severe droughts. When planted mixtures, it should consist of a quick growing legume and a slow growing, but a persistent legume so that a permanent ground cover would be established later (figure 11).

Figure 11: Well grown cover crop

Cover crop management

Once established, the cover crop should not be allowed to grow wild. When the cover crop is too thick and it hampers other field operations, therefore, a mulch roller could be used to control its growth (figure 12): A light harrowing is also possible. Controlled grazing is another method of checking the cover crop. Centrosema and Siratro are particularly grazed by cattle. Pueraria is generally accepted by cattle while Calopogonium seems to be less favored. Buffaloes graze all the species of cover crops. Over grazing by cattle should be avoided as it delays regeneration or sometime causing eradication of cover crop.

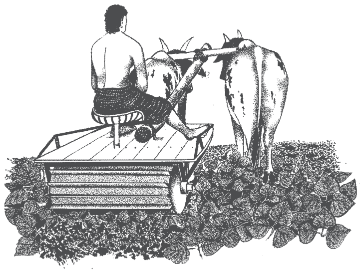

Figure 12: Controlling of cover crops using a roller

When the growth of cover crops is managed with a heavy roller or a disc harrow the entire land should not be covered at once. It is recommended to manage the cover crop with this equipment on alternative rows, because if adverse weather conditions follow the treatments and the cover fail to regenerate. Even in adverse weather conditions, the cover crop in untreated rows spread over the adjacent rows. On slopping lands the cover crop management should be on contours.

The cover crop could be allowed to grow within the manure circle in adult palms. When it is grown in young coconut plantations, it is essential that the cover crop in the area round the young palms up to a distance of 1.8 m (6 feet) is kept free from the cover crop.

Planting and management of bush covers (Gliricidia)

Gliricidia can be planted either from stem cuttings or seeds.

Planting with stem cuttings

Mature cuttings should be taken from stems at least one year old. For live fences, shade trees and live supporters, cuttings of 1.5 m height and 2.5 - 3.0 cm diameter can be planted. The cuttings develop a shallow, laterally-spreading root system. Although establishment by cuttings is convenient, it is suitable mainly for situations where few trees are to be established.

Seed planting

Seed propagation is the most convenient and reliable means of establishment. Gliricidia seedlings develop a deep root system. Pods containing seeds should be collected in April just before maturity as otherwise the pods will shatter resulting in loss of seeds. Seeds can be sown directly, without pretreatment, in the nursery bed, and the seedlings will be ready for planting in 4-6 weeks. The seedlings should be planted with the onset of south-west monsoon. Normally, seedlings can be planted about 15 cm (6 in) deep. In poor soils, addition of 30 g each of Muriate of Potash and Saphos Phosphate to the planting hole will encourage seedling growth.

Planting systems

The most suitable method of planting gliricidia in coconut plantation is along the boundaries as double rows 60 cm (2 feet) apart with inter-row spacing of 60 cm in a triangular system. In the intermediate zone and dry zone, where alley cropping with short - term food crops can be undertaken, Gliricidia can be planted in double rows 3-4m apart with 0.5m within row spacing in the coconut avenue. On any neglected or eroded land where a cash crop cannot be grown, gliricidia can be planted as a green manure crop in double rows, 1-2 m apart with 0.5 m within row spacing in the coconut avenue (figure13).

Figure 13: Double row Gliricidia cultivation

In a Gliricidia cultivation systematic pruning method should be adopted. When the plants are about 1.5 m height about 9-12 months after planting the first cutting can be done. Foliage production can be optimized by harvesting once every three months. For green manure, harvesting can be done at six-monthly intervals. Lopping can be either around 30 cm height above the ground in cropping systems or 1-2 m above ground in live fences and shade trees.

Gliricidia leaves can be used as a green manure. A bout 30 kg of leaves will provide sufficient nitrogen and a part of potash and phosphate for coconut. Whenever gliricidia leaves are available, they could be applied as follows.

- Leaves could be broadcast around the palm and incorporate with a mammoty. The area could then be mulched with coconut fronds.
- Leaves could be left on the surface as mulch. However, when leaves are left on the surface, nitrogen loss is appreciable.

Agronomy Division
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